

Harman Singh

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[LinkedIn](#) [GitHub](#) [LeetCode](#)

Profile

AI/ML Engineer with a focus on building generative AI applications and scalable machine learning pipelines. Experienced in developing RAG-based systems, fine-tuning LLM workflows, and deploying intelligent backend services. Passionate about solving complex technical challenges, backed by a strong foundation in Data Structures, Algorithms, and production-level system design.

Skills

- **Programming:** Python, SQL, C++, JavaScript
- **AI & Data Science** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Feature Engineering, Data Preprocessing
- **Backend & Databases:** FastAPI, REST APIs, API Integration, SQLite, SQLAlchemy
- **Generative AI:** LangChain, LangGraph, LangSmith, RAG Pipelines, Hugging Face, Prompt Engineering
- **Web & Tools:** HTML5, CSS, GitHub
- **Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks

Projects

- AI Document Assistant** GitHub
Engineered a RAG-based document retrieval app using FastAPI and LangChain for context-aware semantic search. Implemented Hugging Face embeddings and FAISS vector storage to optimize retrieval accuracy and response latency. Designed scalable asynchronous backend APIs to streamline data ingestion workflows.
- Kepler Planetary Signal Review System** Live Demo
Developed a full-stack signal classification platform; achieved 89 % recall in signal detection. Automated classification workflows, reducing manual workload by 70 % via an XGBoost-backed engine. Architected secure, modular RESTful APIs with JWT authentication and deployed on Railway/Vercel.
- Telco Customer Churn Prediction System** GitHub
Built an end-to-end ML pipeline, from feature engineering to model evaluation and deployment. Integrated SHAP explainability to provide transparent, interpretable model insights for stakeholders.

Experience

- Python with ML Intern (Project-Based), YBI Foundation** Mar 2024 – Apr 2024
Worked on machine learning workflows, debugging, model evaluation, data preprocessing, and software implementation tasks.
- Python Intern (Project-Based), CodSoft** Dec 2023 – Jan 2024
Built Python applications and strengthened software development, testing, and analytical problem-solving skills.

Education

- B.Tech in Artificial Intelligence & Machine Learning** 2022 – 2026
Guru Tegh Bahadur Institute of Technology
CGPA: 8.8

Achievements

Solved 1500+ DSA problems across coding platforms.
Built multiple backend and AI-powered applications involving APIs, machine learning, Generative AI, and deployment workflows.